

Module-1

Overview of Data Centers

Data Centres Defined, Data Centre Goals, Data Centre Facilities, Roles of Data Centres in the Enterprise, Roles of Data Centres in the Service Provider Environment, Application Architecture Models. The Client/Server Model and Its Evolution, The n Tier Model, Multitier Architecture Application Environment, Data Centre Architecture.

Data Centres Defined:

A data center (sometimes spelled datacenter) is a centralized repository, either physical or virtual, for the storage, management, and dissemination of data and information organized around a particular body of knowledge or pertaining to a particular business.

A datacenter is a facility used to house computer systems and associated components, such as telecommunications and storage systems. It generally includes redundant or backup power supplies, redundant data communications connections, environmental controls (e.g., air conditioning, fire suppression) and various security devices. Large data centres are industrial scale operations using as much electricity as a small town.

A data center, or datacenter, often requires extensive redundant or backup power supply systems, cooling systems, redundant networking connections and policy-based security systems for running the enterprise's core applications. Data center management involves ensuring the reliability of both the connections to the data center as well as the mission-critical information contained within the data center's storage. It also entails efficiently placing application workloads on the most cost-effective compute resource available.

Data Centre Goals

Business goals generate a number of information technology (IT) initiatives:

- Business continuance
- Increased security in the Data Center
- Application, server, and Data Center consolidation
- Integration of applications whether client/server and multitier (n-tier), or web
- services-related applications
- Storage consolidation.

These IT initiatives are a combination of the need to address short-term problems and establishing a long-term strategic direction, all of which require an architectural approach to avoid unnecessary instability if the Data Centre network is not flexible enough to accommodate future changes.

The design criteria are

- Availability
- Scalability
- Security
- Performance
- Manageability.

These design criteria are applied to these distinct functional areas of a Data Center

network:

- Infrastructure services—Routing, switching, and server-farm architecture
- Application services—Load balancing, Secure Socket Layer (SSL) offloading, and caching
- Security services—Packet filtering and inspection, intrusion detection, and intrusion prevention
- Storage services—SAN architecture, Fibre Channel switching, backup, and archival
- Business continuance—SAN extension, site selection, and Data Center interconnectivity.

Roles of Data Centers in the Enterprise- Diagram

The building blocks of this typical enterprise network include

- Campus network
- Private WAN
- Remote access
- Internet server farm:
- Extranet server farm
- Intranet server farm

Enterprise applications typically focus on one of the following major business areas:

- Customer relationship management (CRM)
- Enterprise resource planning (ERP)
- Supply chain management (SCM)
- Sales force automation (SFA)
- Order processing
- E-commerce

Roles of Data Centers in the Service Provider Environment

- Data Centers in service provider (SP) environments, known as Internet Data Centers (IDCs), unlike in enterprise environments, are the source of revenue that supports collocated server farms for enterprise customers.
- The SP Data Center is a service-oriented environment built to house, or host , an enterprise customer's application environment under tightly controlled Service Level agreements for uptime and availability.
- Enterprises also build IDCs when the sole reason for the Data Center is to support Internet-facing applications.

The IDCs are separated from the SP internal Data Centers that support the internal business applications environments.

- Whether built for internal facing or collocated applications, application environments follow specific application architectural models such as the classic client/server or the n-tier model.

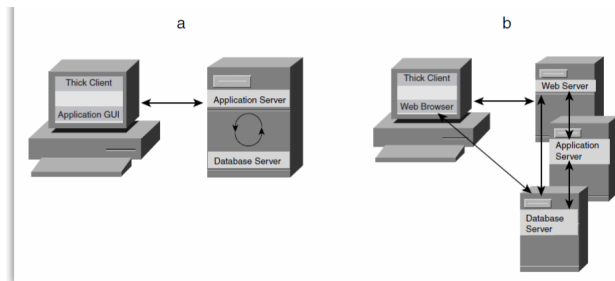
Application Architecture Models

Application architectures are constantly evolving, adapting to new requirements, and using new technologies.

- The most pervasive models are the client/server and n-tier models that refer to how applications use the functional elements of communication exchange.
- The client/server model, in fact, has evolved to the n-tier model, which most enterprise software application vendors currently use in application architectures.
- This section introduces both models and the evolutionary steps from client/server to the n-tier model.

The Client/Server Model and Its Evolution

The classic client/server model describes the communication between an application and a user through the use of a server and a client. The classic client/server model consists of the following:



- A thick client that provides a graphical user interface (GUI) on top of an application or business logic where some processing occurs.
- A server where the remaining business logic resides.

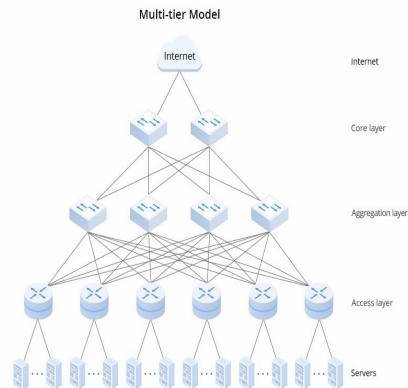
Three-tier or Multi-tier Model

The multi-tier architecture has been the most commonly deployed model of data center architecture used in the enterprise data center, consisting of core, aggregation, and access layers.

Data center core layer: It provides a fabric for high-speed packet switching between multiple aggregation modules and connectivity to multiple aggregation modules.

Data center aggregation layer: It supports functions like service module integration, layer 2 domain definitions, spanning tree processing, and default gateway redundancy.

Data center access layer: It provides the physical level attachment to the server resources and operates in layer 2 or layer 3 modes. What's more, it plays an important role in meeting particular server requirements such as NIC teaming, clustering, and



broadcast containment.

The n-Tier Model

The n-tier model uses a thin client and a web browser to access the data in many different ways.



- The server side of the n-tier model is divided into distinct functional areas that include the web, application, and database servers.
- The n-tier model relies on a standard web architecture where the web browser formats and presents the information received from the web server.

a. Data Center Core Layer

The data center core layer provides a fabric for high-speed packet switching between multiple aggregation modules. This layer serves as the gateway to the campus core where other modules connect, including, for example, the extranet, WAN, and Internet edge. All links connecting the data center core are terminated at Layer 3 and typically use 10 GigE interfaces for supporting a high level of throughput, performance, and to meet oversubscription levels.

b. Data Center Aggregation Layer

The aggregation layer, with many access layer uplinks connected to it, has the primary responsibility of aggregating the thousands of sessions leaving and entering the data center. The aggregation switches must be capable of supporting many 10 GigE and GigE interconnects while providing a high-speed switching fabric with a high forwarding rate. The aggregation layer also provides value-added services, such as server load balancing, firewalling, and SSL offloading to the servers across the access layer switches.

c. Data Center Access Layer

The access layer provides the physical level attachment to the server resources, and operates in Layer 2 or Layer 3 modes. The mode plays a critical role in meeting particular

server requirements such as NIC teaming, clustering, and broadcast containment. The access layer is the first oversubscription point in the data center because it aggregates the server traffic onto Gigabit EtherChannel or 10 GigE/10 Gigabit EtherChannel uplinks to the aggregation layer. Spanning tree or Layer 3 routing protocols are extended from the aggregation layer into the access layer, depending on which access layer model is used. Cisco recommends implementing access layer switches logically paired in groups of two to support server redundant connections or to support diverse connections for production, backup, and management Ethernet interfaces.

Multitier Architecture Application Environment

Multitier architectures refer to the Data Center server farms supporting applications that provide a logical and physical separation between various application functions, such as web, application, and database (n-tier model).

- The network architecture is then dictated by the requirements of applications in use and their specific availability, scalability, and security and management goals.
- For each server-side tier, there is a one-to-one mapping to a network segment that supports the specific application function and its requirements. Because the resulting network segments are closely aligned with the tiered applications, they are described in reference to the different application tiers.

Data Center Architecture

A data center is a physical facility that supports an enterprise's computing activities, which realizes centralized processing, storage, transmission, exchange, and management of information. Data center architecture, as an architectural design that establishes connections between switches and servers, is typically created during the data center design and construction phases. Besides, it designates the way that the server, storage networking, racks, and other data center resources will be placed and also addresses the interconnection of these devices.

Typical Composition of Data Center Architecture

The data center architecture mainly consists of three parts: data center network, security, and computing architecture. Apart from these three, there are some other data center architectures, such as data center physical architecture and data center information architecture. The following will mainly interpret three typical compositions of it.

Data center network architecture: Data center network (DCN) is an arrangement of network devices that interconnect all data center resources together, which has always been a key research area for Internet companies and large cloud computing companies. Therefore, data center network architecture plays a vital role in data center architecture. It usually consists of switches and routers in a two- or three-level hierarchy. For example, three-tier DCN, fat tree DCN, DCell, and other types. Besides, the scale, scalability, robustness, and reliability of the data center network architecture have always been the focus of attention.

Data center security architecture: Data center security refers to physical practices and virtual technologies for protecting data centers from threats, attacks, and unauthorized

access. Data center security architecture mainly includes two aspects: physical security and software security. Enterprises can protect data centers from attack by setting up a strong firewall between the external traffic and the internal network.

Data center computing architecture: Data center computing architecture is one of the most important parts of data center architecture, which plays a huge role in the high-efficient use of resources, reducing capital expenditure (CAPEX) costs, and rapid deployment and scalability. Here we will elaborate on the data center network architecture in detail.

The Internet Edge provides the connectivity from the enterprise to the Internet and its associated redundancy and security functions, such as the following:

- Redundant connections to different service providers
- External and internal routing through exterior border gateway protocol (EBGP) and interior border gateway protocol (IBGP)
- Edge security to control access from the Internet
- Control for access to the Internet from the enterprise clients.

Data Center Facilities

These facilities are likely to support a high concentration of server resources and network infrastructure.

The demands posed by these resources, coupled with the business criticality of the applications, create the need to address the following areas:

- Power capacity
- Cooling capacity
- Cabling
- Temperature and humidity controls
- Fire and smoke systems
- Physical security: restricted access and surveillance systems
- Rack space and raised floors.

Answer the Questions given below

1. Explain the Data Center Architecture with neat diagram.
2. Discuss the Client-Server Model.
3. List the roles of Data Centres in the Enterprise
4. With neat diagram explain n-tier architecture.
5. Discuss the Application Architecture Models.
6. Explain the goal of Data Center.
7. Discuss the Data Center Facilities in detail.
8. Draw the neat diagram of Data Center Architecture and explain all the three layers and its functions.
9. Discuss the Data Center Facilities in detail.